

Wireless Communications - NUIST

Engineering Technology

May, 2026

Wireless Communications - NUIST (A35630)

Short Title:	Wireless Communications (NUIST)	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Discipline Specific Technologies	
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Credits	5	Level: Postgraduate

Description of Module / Aims

The course covers software defined radio, where the transmitter and or receiver modulation and architecture may be dynamically reconfigured. The emphasis is on baseband communications. Optimising signal recovery in a typical high frequency wireless system is also covered. Exploitation of diversity is introduced in a section on multiple input/ multiple output communications. Finally, advanced wireless modulation techniques are explored.

Programmes

COMM-0004	MEng in Electronic Engineering (WD_EELEN_R)	1 / 2 / E
	MEng in Electronic Information Systems (WD_EELEN_RI)	2 / 3 / E
COMM-0004	PG Dip in Engineering in Electronic Engineering (WD_EELEN_G)	1 / 2 / E

Indicative Content

- Baseband Software Defined Radio Transceiver Line-up
- Radio propagation and spatial domain Processing
- Multiple input and multiple output wireless communications and adaptive antennas
- Advanced modulation (Case study: EDGE, UWB)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Design and synthesis an entire flexible transmit-receive baseband architecture.
2. Circumvent channel distortions and unwanted additive interference, to optimise SNR for signal recovery and maximum data throughout.
3. Describe and analyse a layered space-time environment to achieve ultra high density data throughout, employed in a fixed wireless configuration with smart antennas.
4. Explain and compare modern advanced modulation techniques as used in narrowband and wideband systems.
5. Demonstrate practical operation of experimental apparatus/simulation for each laboratory session and mini-project. The student should show a familiarity with the experiment/apparatus & a clear understanding of the procedure/topic under consideration.

Learning and Teaching Methods

- Lecture
- Directed Research
- Moodle

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	20%	3,4
Project	30%	1,2,5
Lab Report	30%	3,4
Practical	20%	1,2,5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	6	
Independent Learning	105	

Essential Material

- Feher, K. *Digital Communications - Satellite / Earth Station Engineering*. New Jersey: Prentice Hall, 1983.
- Feher, K. *Wireless Digital Communications - Modulation and Spread Spectrum Applications*. New Jersey: Prentice Hall, 1995.
- Jakes, W. *Microwave Mobile Communications*. New Jersey: IEEE-Press, , 1974.
- Proakis, J.G. *Digital Communications*. : McGraw Hill, 1989.
- Skiar, B. *Digital Communications - Fundamental Applications*. New Jersey: Prentice Hall, 1988.
- Steele, R. *Mobile Radio Communications*. : Prentech, 1992.